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Towards autonomous greenhouses solar-powered

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Abstract

Fuzzy logic controls have been used to develop intelligent control systems in greenhouses, leading to promote a comfortable microclimate for plant growth while saving energy and water resources. Solar energy is an effective solution to extend the use of these intelligent systems in isolated areas. In fact, this paper presents a practical solution based on the power-aware design for controlling the indoor environment of solar greenhouses. The system consists of feedback fuzzy logic controllers that log key field parameters using specific sensors and performs control actions for the environmental temperature, soil moisture, and relative humidity. With regard to the battery-coupled architecture of the solar system, the decision level is designed to consider the availability of operational resources before acting. Keeping in mind the constraint of low-cost, we propose a flexible design of an embedded system able to support remote management of the indoor environment key-parameters. The proposed control system was validated experimentally. The efficiency of the system was assessed by demonstrating significant energy and water savings as well as protection of solar system components from dry running issues

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1. Introduction

High-tech greenhouses based on new modern information technology, have focused on the control of indoor environments since they adapt to climatic conditions changes, which leads to optimize crop growth conditions and the growing season [1]. In addition, they provide efficient irrigation management by avoiding unnecessary pumping and water leakage, which reduces operating costs and increases production efficiency [2], [3]. However, implementing an intelligent control system in a greenhouse is a complex process because of the number of variables involved and the strong dependency among them. To meet these requirements, different control strategies based on complex algorithms of artificial intelligence have been discussed in the literature [4]. In fact, fuzzy logic systems (FLS) have achieved significant results in the area of climate control for protected agriculture. Many control techniques based on FLS are often used for this type of system [5]–[7]. In particular, automated irrigation systems based on the fuzzy logic approach are able to automatically provide the exact amount of water required for growth according to crop needs from soil moisture levels through appropriate sensors [8]–[12].

Recently, renewable energy resources have been incorporated as a source of power for automated systems, which extend their use even in isolated areas where there is no access to conventional electricity. Particularly, solar-powered systems support long power system autonomy by taking the advantage of the microgrid technology and improving the energy use efficiency [13]–[16]. However, the integration of solar energy as a source of energy in the context of an autonomous greenhouse is accompanied by significant changes in decision-making. For instance, besides scheduling the activation of actuators in order to meet only crop needs, the new design must take into consideration the availability of operational sources before acting.

Several works have focused on new approaches for implementing this automated system as economically cheap solutions in order to lower their costs to a level where growers can actually use them. Currently, Raspberry Pi is becoming the traditional system of many works [17]–[19]. This card has more advantages like a better cost/benefit ratio, a larger community of users, a standard programming language and communication, many input/output pins and a graphical interface.

In this work, a fully automated solar-powered system for monitoring and control of a greenhouse environment is proposed. For this reason, a new multithreaded and power-aware design concept using open source and low-cost features has been used to integrate a smart fuzzy logic-based control system into the Raspberry Pi. The use of advanced programming features, namely multithreading, enable the integration of concurrent services within the system.

2. System design

In this study, an embedded control system is introduced to control the temperature, relative humidity, and soil moisture in a solar greenhouse as shown in Fig. 1.(b). It consists of four main subsystems: sensing, climate computer, actuators, and photovoltaic (PV) system.

The sensing unit consists of several specific field sensors which log the external data to the acquiring card. The sensors of the climate are the temperature and relative humidity sensor (DHT22) and the soil moisture sensor (VH400), while the sensors of operational resources are the water level sensor (HC-SR04) and the multimeter MX5060 for batteries voltage.

The actuators used to regulate the environment in the greenhouse are ventilation, humidification, heating, and irrigation systems. Actuator systems are mainly composed of electrical devices, specifically by electrical DC /AC motors or pumps. The forced ventilation consists of an AC fan responsible for generating an airflow that enters the greenhouse, as well as an additional extractor which is used to extract hot air. The humidification system consists of an AC air compressor. However, in order to decrease the relative humidity, a system of forced ventilation was used to generate a decrease in the relative humidity by raising the temperature and exchanging the air inside the greenhouse. The last actuator, which is used to control the soil moisture, is a solar pump of 24 V DC.

The Raspberry Pi (RPI) microcomputer is used as kind of a climate computer to host the intelligent system. The card runs with the Linux-based embedded devices [18], which is an open source operating system providing additional advantages to use several modern programming languages such as C, C++, Python, or Java. In this study, Java Platform is adopted to develop the software part and the Java Virtual Machine (JVM) is used as the runtime engine.

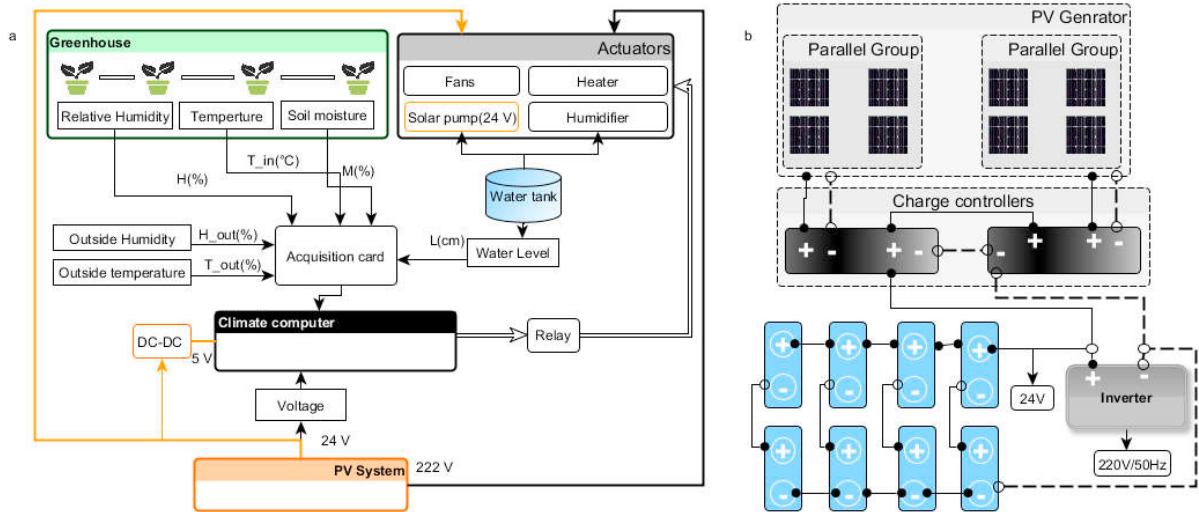


Fig. 1. (a) Schematic diagram of the solar greenhouse system, (b) Schematic diagram of the PV system.

The PV generator is composed of two parallel sets of series and parallel PV arrays where each individual set is linked to a charge controller as shown in Fig. 1.(b). Each charge controller is used with the purpose to adjust the current and voltage of the solar panels in order to charge batteries. The storage bank is composed of eight lead-acid batteries of 100 Ah and 12 V that used in parallel and series to obtain 400 Ah and 24 V. A single-phase 24 V/220 V inverter is used to generate an alternative voltage. Therefore, the power system is able to provide constant voltage and sufficient current. Indeed, the battery bank with the capacity left Q (AH) is able to supply requested energy to the actuators with continuous current I_{dis} (A) in order to perform irrigation task for duration D in hours which is given by:

$$D = \frac{Q}{I_{dis}} \quad (1)$$

3. Climate computer

The smart system is designed and developed around the Raspberry Pi as shown in Fig. 2. The system consists of three major services which are the Data logger, Control unit, and Task manager, running together as a paralleled thread. Each thread performs a specific task with data exchange with others services.

The data logging service is responsible for collecting external data from the acquisition data card using a serial connection. Available climate data and operational resources are uploaded to the climate computer using serial communications. The temperature sensors, humidity sensors, soil moisture sensor and water level sensor are interfaced with the Arduino Duo board which serves as a receiver. As well, the climate computer can retrieve the current measurement of battery bank voltage using the MX5060 via a USB connection. The collected data is then passed to the control unit for further analysis, stored in the local database, and made available to remote users.

The Control unit service is made of two levels: an operational level that performs the actual control and a strategic level that serves as a kind of supervisor. The operational level consists of three fuzzy logic-based controllers, which take the observations on the greenhouse physics, and returns control variables in the form of actuator commands. However, these operators are dormant components waiting for notification from the supervisor to act. After receiving real-time measurement, the supervisor performs a deep analysis based on this data and the values of set points to decide what kind of actuators to activate. When an action is selected, the supervisor provides the corresponding controller with the appropriate key-input data. For instance, the Humidifier controller uses the outside humidity (H_{out}), inside humidity(H_{in}), inside temperature (T_{in}), and humidity set point (H_{set}) as input parameters to estimate the humidifying time (D_{th}) as the output variable. The temperature controller uses the output temperature

(T_{out}), inside temperature (T_{in}), inside humidity, and temperature setpoint as input parameters to estimate the ventilation time (D_{th}) as the output variable. Whereas, the irrigation controller uses the inside temperature (T_{in}), the soil moisture (SM), and the soil moisture set point (SM_{set}) to estimate the watering time (D_{th}) as the output variable. At the end of the process, the current operation information including the type and estimated duration are passed to the task manager to decide whether to add a new task to the scheduler or to issue a warning message.

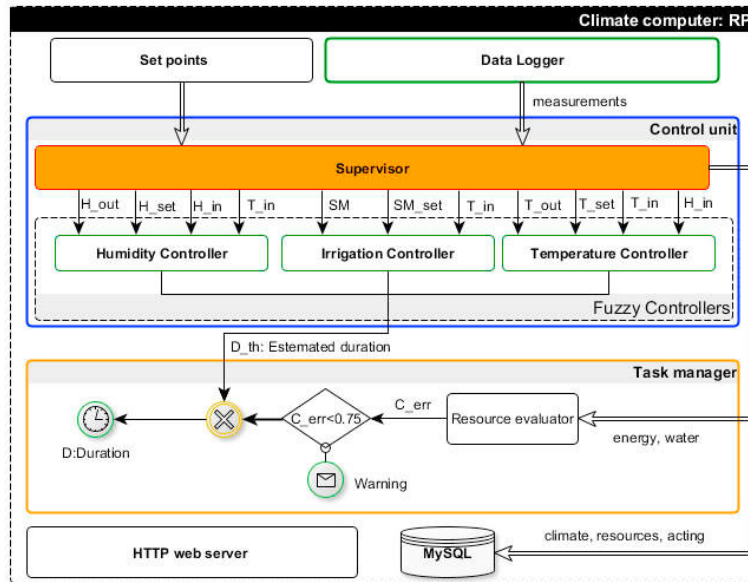


Fig. 2. The architecture of the fuzzy logic-based control system.

The task manager consists of a resource evaluator block capable of estimating the ability of the system to maintain the requested action for the requested duration. For instance, the ventilation and heating require only sufficient power as the only operational acting resource. However, the irrigation and humidification also require the amount of water available in the tank. The coverage error (C_{err}) is the ratio of the requested action duration to the guaranteed acting duration of the stored resources. If the C_{err} parameter is greater than 75%, the task manager issues a warning message with a high priority; otherwise, the task manager schedules a new task to activate the actuator for the calculated duration as follows:

$$D = D_{th} * C_{err} \quad (2)$$

MySQL database is used to store locally the collected data that represent the states of the solar system. The database consists of three main entities, which are measurement, actuator, and configuration. In addition, an HTTP web server is introduced with the purpose of remote management and access to the local control station.

4. Prototype and proof of concept

The greenhouse used as support for this study is an experimental greenhouse located at the laboratory of Electronic, Automatic and Biotechnology of the Faculty of Sciences. The greenhouse is single wall greenhouse in which the sensors, the control system, and the final control actuators were installed. The greenhouse is equipped also with a battery -coupled architecture photovoltaic system as a power supply source as describe shown in Fig. 3.

The control system is developed around The Vert.x tool which is an open source and lightweight platform that helps to set up real-time applications based on reactive micro-services known as Verticles [20]. It is also a polyglot platform with the flexibility to incorporate different programming languages to deploy Verticles. However, in this study, the Vert.x is used as a toolkit running on the Java Virtual Machine (JVM). The most important feature of Vert.x

is its capability of using all the central process units (CPUs) in machines or cores in CPU. This is performed by creating one thread per CPU or core using the Multi-Reactor Pattern. Each thread can distribute messages to multiple Verticles using the Events bus which is an integrated and distributed service for the internal communication purpose. Verticles are event-driven components running only when receiving messages. For instance, the application logic is distributed between several threads running in parallel with usually a synchronized data exchange with other services. On the other hand, the Vert.x provides an execution context for deploying the HTTP server within the application as a single thread or Verticle of the thread rather than operating as a standalone service.



Fig. 3. (a) Single wall greenhouse, (b) Battery coupled PV system.

5. Results and discussion

The developed FLC system is applied to the experimental greenhouse during different periods in order to test all type of actuation of the control system. Indeed, the solar control system is evaluated when using the proposed FLC strategy to face with the unavailability of operational resources when acting. The behavior of the control system to manage the actuator activations, considering the availability of water and power, is presented in Table 1. For each actuator, three activation durations have been introduced namely, requested, guaranteed and actual. Requested durations represent durations that are estimated to activate a specific actuator to satisfy the optimal growing conditions of the crops. Guaranteed durations indicate actual durations provided by the operational resources available to activate a specific actuator according to its operating conditions. Finally, actual durations are the effective durations to activate a specific actuator to meet crop needs while preventing farm facilities from dealing with run dry issues.

Table 1: Control system behavior

Case	Actuator	Actuation durations (H: Mn:S)		Coverage errors (%)	Actual activations (H: Mn:S)
		Requested	Guaranteed		
1	Irrigation	00:10:00	00:11:30	0%	00:10:00
2	Heater	00:06:00	00:11:30	0%	00:06:00
3	Humidifier	00:08:00	00:11:30	0%	00:08:00
4	Irrigation	00:10:00	00:10:00	5%	00:09:30
5	Irrigation	00:10:00	00:05:30	50%	00:05:00
6	Heater	00:06:00	00:06:00	8%	00:05:30
7	Humidifier	00:08:00	00:08:00	6%	00:07:30
8	Humidifier	00:08:00	00:04:00	56%	00:03:30
9	Heater	00:06:00	00:02:30	67%	00:02:00
10	Irrigation	00:10:00	00:01:00	95%	00:00:00
11	Heater	00:06:00	00:01:00	92%	00:00:00
12	Humidifier	00:08:00	00:01:00	94%	00:00:00

As it can be seen:

- When operational resources are sufficiently available, the control system activates the concerned actuator during the requested duration, as in the cases [1-3].
- The control system is able to prevent in advance the shorten in the autonomy of the solar facility and decides to activate the concerned actuator for safe durations that are less than requested ones, as in the cases [4-9].
- When the available resources are very low, it is useless to plan unsuccessful tasks, as in the case [10-12]

6. Conclusion

We have presented a new approach to develop an embedded system based on a multithreaded design to manage and monitor the microclimate and operational resources in a solar greenhouse context. The availability of operational resources was introduced as new key parameters to evaluate acting time when it is needed. The fuzzy logic control system was implemented to automate the activation of the final control elements with particular attention to the effectiveness, water use efficiency, energy use efficiency, and solar components safeness as well. Owing to the multithreading programming feature provided by the reactive tool Vert.x, the control process and an HTTP server were integrated into two parallel tasks working together to provide remote control and online data monitoring of solar greenhouse microclimate in real-time. Also, the Raspberry Pi was selected to embed the smart system and demonstrated its ability to meet the constraints in real-time applications.

The next step will focus on enhancing the system in order to be deployed in large-scale areas. This challenge involves systematically the issues of deploying measurement nodes in large fields.

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